

LEKSHMI J

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PROFESSIONAL SUMMARY

Results-driven **Data Scientist** with hands-on experience in **machine learning, deep learning, NLP, computer vision, and advanced analytics** across diverse domains. Skilled in designing and deploying **end-to-end ML solutions, predictive models, and recommendation systems** that enhance decision-making and business performance. Proficient in **Python, SQL, R, machine learning and deep learning libraries and statistical modelling** with proven ability to work on **large-scale, complex datasets**. Experienced in **feature engineering, data preprocessing, model training, validation, and deployment** with expertise in **cloud-based workflows and data pipelines**. Delivered **production-ready AI/ML applications, automated systems, and interactive dashboards** that improved operational efficiency. Adept at collaborating with cross-functional teams, translating business needs into data-driven strategies, and driving measurable impact through **AI/ML innovation**.

PROFESSIONAL EXPERIENCE

- 1. Data Science Intern | OCEAN AI(MariApps Marine Solutions Pvt. Ltd) | 02/2025 – 06/2025**
 - Built a **deep learning–based crew recommendation system** achieving **79% accuracy**, optimizing **193K+ workforce allocation recommendations** in maritime operations.
 - Developed a **PyTorch encoder-decoder model** trained on **157K records**, reducing prediction error by **19%** and achieving **99.5% similarity** in crew-vessel matching.
 - Designed and deployed a **scalable ETL data pipeline** processing **642+ time-period datasets**, reducing computation time and enabling **real-time decision-making**.
 - Implemented a **FAISS-powered similarity search engine** that automated candidate ranking by experience compatibility and vessel requirements.
 - Reduced **manual crew selection time by 70%**, delivering **40+ relevant candidate suggestions per query** via a live production system.
 - Benchmarked **LightGBM, Neural Collaborative Filtering, and Deep Neural Networks**, selecting the optimal architecture through **rigorous experimentation and validation**.
- 2. Accounting Services Associate I | Conduent Business Services India LLP | 01/2022 – 06/2023**
 - Processed and validated **high-volume financial datasets**, ensuring **data integrity, compliance, and reporting accuracy** across multiple client accounts.
 - Built **dynamic Excel-based dashboards** with advanced formulas and pivot tables, enabling **real-time financial tracking and reporting** for leadership teams.
 - Cleaned, transformed, and analyzed **vendor payment datasets**, improving reporting efficiency and reducing manual errors in **accounts payable workflows**.
 - Trained and mentored **8 team members** in data analysis, reporting automation, and standardized workflows, enhancing **team-wide analytics capability**.
 - Utilized **SAP ERP and Excel** for **data integration, financial modeling, and transaction management** in enterprise-scale reporting systems.

EDUCATION

M.Sc. Data Analytics | Christ University, Bangalore | 2023 – 2025

- CGPA: **9.72/10**

B.Com (Finance) | N.S.S College, Pandalam (University of Kerala) | 2018 – 2021

- CGPA: **8.69/10**

TECHNICAL SKILLS

- **Programming & Query Languages:** Python (Pandas, NumPy, Scikit-Learn, Matplotlib, Seaborn, OOP), SQL (MySQL), R, Git
- **Machine Learning & Statistics:** Supervised & Unsupervised Learning (Regression, Classification, Clustering, PCA), Predictive Modeling, Feature Engineering, Data Preprocessing, Statistical Analysis, Hypothesis Testing, Model Evaluation & Validation
- **Deep Learning & Neural Networks:** PyTorch, TensorFlow/Keras, CNNs, LSTM, MLP, Encoder-Decoder Models, Time Series Forecasting
- **Natural Language Processing & Computer Vision:** NLTK, spaCy, Sentiment Analysis, Text Preprocessing, BLEU/WER Evaluation, Image Classification, OpenCV
- **Generative AI & LLMs:** Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), Fine-tuning Models
- **Data Analysis & Visualization:** Power BI, Tableau, Excel (Pivot Tables, Advanced Functions), Power Query, EDA, Data Cleansing, Data Storytelling
- **Web Development & Deployment:** Streamlit, Gradio, FastAPI, REST APIs, Model Deployment & Integration
- **Databases & Big Data:** MySQL, MongoDB (Basic), Hadoop, Spark, FAISS (Similarity Search)
- **MLOps:** MLflow, DVC, Docker, CI/CD (GitHub Actions), Cloud (AWS – Basic)
- **Tools & Platforms:** Jupyter Notebook, MS Office Suite, Interactive Dashboards, Version Control

PROJECTS AND ACADEMIC WORKS

❖ CNN Image Classification with Transfer Learning

- Built an end-to-end image classifier on **25K Kaggle images** using **MobileNetV2** and **TensorFlow/Keras**.
- Designed preprocessing pipeline (resizing, normalization, 80/20 split).
- Fine-tuned MobileNetV2 with **Adam optimizer**, achieving **98% accuracy in 5 epochs**.

❖ Automated Waste Classification System (Deep Learning & Computer Vision)

- Developed a **5-class image classifier** using **InceptionV3 + transfer learning**, with custom regularization layers to reduce overfitting.
- Collected & augmented dataset via **RoboFlow** (rotation, flip, brightness), improving model robustness.
- Deployed real-time **Gradio web app**, achieving **71.1% accuracy**, showcasing full **ML pipeline deployment** and sustainability applications.

❖ Cybersecurity Attack Monitoring Dashboard (Power BI & Python)

- Built dashboard to track 20K+ cyber incidents, analyzing logs and network activity for threat patterns. Designed interactive visuals to highlight risk zones, trends, and anomalies with 90% pattern accuracy. Identified top attack sources and high-risk platforms; delivered 17 key insights to reduce future threats

❖ YouTube Sentiment Insights | End-to-End MLOps Project

- Engineered automated sentiment analysis for 37K+ YouTube comments using LightGBM (65% accuracy) with advanced NLP preprocessing (tokenization, lemmatization, TF-IDF, trigrams, SMOTE) and experiment tracking of 10+ models via MLflow.
- Developed Chrome extension and Flask REST API for real-time comment analysis, interactive dashboards (charts, word clouds, trends), and seamless model inference.
- Implemented full MLOps workflow: **DVC pipelines**, **Docker containerization**, **CI/CD via GitHub Actions**, and **AWS deployment (EC2, S3, ECR)**, enabling reproducible, scalable, production-ready end-to-end solution.

ACHIEVEMENTS

- Achieved **perfect 4.0/4.0 CGPA** in the second trimester of **M.Sc. Data Analytics** program.
- Awarded **Honor Roll Award** for outstanding **team performance and project leadership** at Conduent Business Services India LLP.
- Secured **1st position in B. Com (Finance)** degree examinations at the college level.
- Received **Certificate of Merit** for achieving a **perfect 10/10 CGPA** in Class 10 and ranked **1st in Class XII** examinations at school level.

CERTIFICATIONS

- **Machine Learning Specialization** – DeepLearning.AI
- **Google Data Analytics Professional Certificate** – Google